

FlowTrail

Developer Documentation

Full-Stack Personal Productivity PWA

Built by Harish Sivakumar · May 2026

github.com/harishsivakumarjs/flowtrail · flowtrail-delta.vercel.app

1. Project Overview

FlowTrail is a full-stack personal productivity Progressive Web App (PWA) built entirely from scratch. It combines habit tracking, task management, journaling, Google Calendar integration, focus mode, analytics, and gamification into a single workspace that syncs in real time across all devices.

The project started with a simple idea: most productivity apps do one thing well but nothing does everything. FlowTrail was built to solve that problem and grew into a production-grade application.

Live URL: <https://flowtrail-delta.vercel.app> GitHub:
<https://github.com/harishsivakumarjs/flowtrail>

1.1 Core Goals

- Offline-first — app works without internet, syncs when reconnected
 - Real-time — changes on one device appear on all others within 2 seconds
 - Mobile-first — installable as a PWA on Android, thumb-friendly navigation
 - Open source — clean codebase for portfolio and interview demos
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2. Tech Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite	Component-based UI, fast HMR dev server
Styling	TailwindCSS	Utility-first CSS with CSS variables for theming
State	Zustand	Global state with localStorage persistence
Offline DB	Dexie.js	IndexedDB wrapper for offline-first data storage
Backend	Supabase	PostgreSQL + Auth + Real-time WebSocket subscriptions
Auth	Supabase Auth	Email/password + Google OAuth (PKCE flow)
Editor	Tiptap	Rich text editor (ProseMirror-based) for journal
Charts	Recharts	Pie, bar, and line charts for analytics pages
Routing	React Router v6	Client-side SPA routing with nested routes
Dates	date-fns	Lightweight date manipulation and formatting
PWA	vite-plugin-pwa	Service worker + Web App Manifest generation
Calendar	Google Calendar API	Read and write Google Calendar events
OAuth	Google Identity Services	Popup OAuth flow for Calendar access token
AI	Anthropic Claude API	AI task planning via Vercel Edge Function
Deploy	Vercel	Frontend hosting + serverless Edge Functions

3. Architecture

3.1 Project Folder Structure

```
flowtrail/
├── api/
│   └── plan.js          # Vercel Edge Function — AI proxy (avoids CORS)
├── public/
│   ├── icon-192.png     # PWA icons for Android install
│   └── icon-512.png
├── src/
│   ├── components/layout/
│   │   ├── AppShell.jsx # Main wrapper with sidebar + bottom nav
│   │   ├── Sidebar.jsx  # Desktop navigation with NavLink active states
│   │   └── BottomNav.jsx # Mobile 5-tab bar (Home/Habits/Tasks/Calendar/More)
│   ├── components/ui/
│   │   ├── XPBar.jsx    # XP progress bar with level + toast on gain
│   │   ├── FocusScore.jsx # Live focus score badge + warning toasts
│   │   ├── PlanMyDay.jsx # AI planner modal
│   │   └── Modal.jsx     # Reusable modal wrapper
│   ├── hooks/
│   │   ├── useHabits.js  # Habit CRUD + XP rewards + Google Calendar sync
│   │   ├── useTasks.js   # Task CRUD + notifications + XP rewards
│   │   └── useJournal.js # Journal CRUD — multiple entries per day
│   ├── lib/
│   │   ├── supabase.js   # Supabase client with PKCE auth config
│   │   ├── db.js         # Dexie schema — v1→v2→v3 migrations
│   │   ├── sync.js       # Bi-directional sync engine with echo prevention
│   │   ├── gamificationSync.js # Push/pull XP+badges to Supabase profiles
│   │   ├── googleCalendar.js # Google Calendar API (fetch/create/update/delete)
│   │   └── notifications.js # Web Notifications API helpers
│   ├── pages/
│   │   ├── Landing.jsx   # Marketing page + auth (email + Google)
│   │   ├── AuthCallback.jsx # OAuth callback handler
│   │   ├── Dashboard.jsx # Home — stats, heatmap, habits, tasks
│   │   ├── Habits.jsx    # Month grid + pie chart + bar chart
│   │   ├── Tasks.jsx     # Task list — today/upcoming/all
│   │   ├── Calendar.jsx  # Month calendar + Google Calendar sync
│   │   ├── Journal.jsx   # Multi-entry rich text journal
│   │   ├── Analytics.jsx # Weekly score + trends + badges
│   │   ├── Focus.jsx     # Pomodoro + web blocker + intent gate
│   │   └── Settings.jsx  # Theme + export + about
│   ├── store/
│   │   ├── appStore.js   # Theme + user session (persisted)
│   │   ├── gamificationStore.js # XP + levels + badges (persisted)
│   │   ├── dopamineStore.js # Focus score + distraction tracking
│   │   └── focusStore.js  # Blocked sites + sessions (persisted)
├── supabase/
│   └── schema.sql        # Full DB schema + all migrations
├── vercel.json           # SPA rewrite — all routes → index.html
└── vite.config.js        # PWA manifest config
```

3.2 Sync Flow

FlowTrail uses an offline-first sync architecture. Every write goes to IndexedDB first for instant UI feedback, then pushes to Supabase in the background.

```
User action (e.g. complete a habit)
↓
markLocalWrite(id) — register ID to ignore own real-time echo
↓
db.habit_logs.put(record) — write to Dexie (IndexedDB) instantly
↓
upsertToCloud(table, record) — POST to Supabase REST API
↓
Supabase broadcasts via WebSocket (Postgres Changes subscription)
↓
Other devices: onAuthStateChange listener → db[table].put(payload.new)
↓
useLiveQuery() triggers re-render → UI updates on all devices
```

3.3 Database Schema

Table	Key Columns	Notes
profiles	id, xp, earned_badges, total_habits_done	Gamification sync — one row per user
habits	id, user_id, name, icon, color, goal_value, archived	goal_value = user-defined monthly target
habit_logs	id, habit_id, user_id, log_date, completed	One row per habit per day per completion
tasks	id, user_id, title, priority, status, due_date, due_time	due_time added in Dexie v3 migration
journal_entries	id, user_id, entry_date, title, content, word_count	No UNIQUE constraint — multiple/day allowed
sleep_logs	id, user_id, log_date, hours	Logged from dashboard sleep input

4. Features — Technical Details

4.1 Habit Tracking

- Month-view grid — 31 day cells per habit rendered as clickable div elements

- Synced horizontal scroll — all rows + header share a single onScroll handler via `querySelectorAll(".habit-scroll")`
- Auto-scrolls to today on mount using `useEffect` — `scrollLeft = (todayDay - 4) * CELL_WIDTH`
- Streak = consecutive days from today going backwards, computed each render
- Custom monthly goal (`goal_value`) — pie chart compares done vs goal, bar chart shows % of goal
- Habit toggle calls `markLocalWrite(id)` before update to prevent real-time echo

4.2 Task Management

- `due_time` added as new field in Dexie v3 schema (non-destructive, no data migration)
- Future task warning — if `due_date > today` and status is pending, shows 3-option modal: complete anyway / delete / keep
- Upcoming view groups by date with day-of-week headers, sorted by date then by `PRIORITY_ORDER = {high:0, medium:1, low:2}`
- 30-minute browser notification uses `setTimeout(fn, dueDateTime - Date.now() - 30*60*1000)`
- Auto-creates Google Calendar event if `isCalendarConnected()` returns true when `addTask()` is called

4.3 Journal

- Old Supabase schema had `UNIQUE(user_id, entry_date)` constraint — dropped with `ALTER TABLE` migration
- Each entry has its own UUID primary key — unlimited entries per day per user
- Tiptap editor uses StarterKit + Placeholder extensions
- Auto-save — `setInterval` every 30s calls `updateJournalEntry()` if editor is not empty
- Tab strip renders `todayEntries` array as buttons — clicking loads that entry into editor
- `createJournalEntry()` vs `updateJournalEntry()` determined by whether `activeEntryId` is set

4.4 Google Calendar Integration

- Uses Google Identity Services (GIS) popup — entirely separate OAuth flow from Supabase auth
- Access token stored in `localStorage` (`gcal_token` + `gcal_expiry`) — `isCalendarConnected()` validates both
- Calendar page auto-reconnects on mount if token exists and is not expired

- FlowTrail-created events excluded from display — description contains "[FlowTrail]" marker
- Events filtered after fetch so the calendar grid shows only genuine external Google events
- Import modal — user selects Google events → addTask() called for each → they appear in Tasks page
- 30-minute notifications scheduled on Calendar page load for today's Google events

4.5 Focus Mode

- Pomodoro timer — circular SVG ring using strokeDashoffset = circumference * (1 - pct/100)
- Web blocker works inside FlowTrail only — PWA cannot intercept system-wide browser traffic
- Link checker: paste URL → extract domain → check against blockedSites array → apply rules
- Daily limits tracked in focusStore.siteUsage[site][today] (minutes used, persisted)
- Delay timer — setInterval decrements seconds, opens URL when reaches 0
- Intent gate — logs site + reason to intentLog array in focusStore
- Friction ladder — attempts[site] counter, DELAYS[min(count, 3)] selects wait time

4.6 Gamification

- Zustand gamificationStore persisted to localStorage under key "flowtrail-gamification"
- addXP() uses set(s => ...) functional update to avoid stale state
- recordHabitDone(), recordTaskDone(), recordSession() call addXP() then check badge thresholds
- pushGamification() called immediately after XP gain — upserts to Supabase profiles table
- pullGamification() on login — Math.max() ensures XP never goes backward when merging devices
- XPBar toast — useEffect watches lastXPGain.time, renders XPToast with 2-second auto-dismiss

4.7 AI Planner

- Browser cannot call Anthropic API directly — blocked by CORS policy

- Vercel Edge Function at /api/plan.js acts as thin proxy — receives goal, calls Claude, returns tasks
- Uses claude-haiku model for speed and low cost
- Prompt instructs model to return ONLY a JSON array — no markdown, no preamble
- Response parsed with regex `/[\s*\S*]/` to extract JSON array safely
- PlanMyDay.jsx calls /api/plan on Vercel — does not work on localhost (edge functions are Vercel-only)

5. Problems Faced & Solutions

Problem	Root Cause	Solution
Habit toggle broken	Dexie compound index query failing silently	Switched to <code>.where("habit_id").equals(id).filter(l => l.log_date === today)</code>
All IDs were integers, Supabase rejected them	Dexie v1 used ++id (auto-increment)	Dexie v2 migration: clear all tables, switch to id (string), use <code>crypto.randomUUID()</code>
Deleted tasks reappeared after refresh	<code>syncFromCloud</code> re-pulled deleted records from Supabase	Added <code>deletedIds</code> Set — delete from Supabase first, filter in <code>syncFromCloud</code>
Habit toggle flickered (toggled twice)	Real-time subscription echoed own write back	<code>markLocalWrite(id)</code> ignores real-time events for own IDs for 3 seconds
F key in journal triggered Focus shortcut	Keyboard handler only checked <code>tagName</code> , not <code>contenteditable</code>	Added check for <code>.ProseMirror</code> and <code>[contenteditable]</code> ancestor elements
CORS blocked Anthropic API from browser	Anthropic API does not allow browser requests	Vercel Edge Function /api/plan proxies the request server-side
Magic link redirected to localhost on mobile	Supabase Site URL was set to localhost	Updated Supabase Authentication > URL Configuration to Vercel URL
Email rate limit exceeded	Supabase free tier: 3 magic link emails per hour	Switched to email + password auth — no emails sent on login
Google OAuth Error 403 access_denied	App in Testing mode, email not in test users list	Added email in Google Cloud Console > Audience > Test users

PWA not installing fullscreen	Icon URLs pointed to external CDN returning 404	Created icon-192.png and icon-512.png locally in /public folder
Journal could not save multiple entries per day	Supabase had UNIQUE(user_id, entry_date) constraint	ALTER TABLE journal_entries DROP CONSTRAINT ... — run in SQL Editor
XP not increasing on habit completion	Code called recordHabitCheck() — function does not exist	Fixed to recordHabitDone() which exists in gamificationStore
Gamification not syncing across devices	XP only in Zustand localStorage, not in Supabase	Added xp/earned_badges columns to profiles table, added push/pull functions
Bottom nav always showed Home as active	All NavLinks matched / prefix — no end prop	Added end={true} only on the Home NavLink to require exact path match
Calendar showed duplicate tasks (FlowTrail + Google)	addTask() auto-created Google event, both showed in calendar	Filter Google events where description includes [FlowTrail before rendering
Habit grid date header not scrolling with rows	Header had overflow-x: hidden, could not scroll	Changed to overflow-x: auto with pointerEvents: none — rows drive scroll via headerRef.scrollLeft
Google Calendar token expired after 1 hour	GIS tokens have 1 hour expiry by default	Store expiry timestamp in localStorage, isCalendarConnected() validates both token and expiry

6. Deployment

6.1 Vercel Setup

- Connect GitHub repo to Vercel — auto-deploys on every git push to main
- vercel.json contains SPA rewrite rule: all routes → /index.html
- Edge functions placed in /api folder — automatically detected by Vercel

Environment Variable	Where to get it	Used in
VITE_SUPABASE_URL	Supabase > Settings > API	Frontend (Supabase client)
VITE_SUPABASE_ANON_KEY	Supabase > Settings > API	Frontend (Supabase client)
VITE_GOOGLE_CLIENT_ID	Google Cloud Console > Credentials	Frontend (GIS OAuth)

ANTHROPIC_API_KEY	console.anthropic.com > API Keys	Edge function only — never frontend
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6.2 Supabase Setup

- Enable Email provider in Authentication > Providers > Email (disable email confirmation for easy testing)
- Enable Google provider — paste Client ID and Client Secret from Google Cloud Console
- Add Vercel URL and localhost to Authentication > URL Configuration > Redirect URLs
- Run schema.sql in SQL Editor to create all tables with Row Level Security
- Enable Realtime on all tables: ALTER PUBLICATION supabase_realtime ADD TABLE ...

6.3 Google Cloud Console Setup

- Create OAuth 2.0 client (Web application type)
- Authorised JavaScript origins: https://flowtrail-delta.vercel.app + http://localhost:5173
- Authorised redirect URIs: https://[project].supabase.co/auth/v1/callback + Vercel URL + localhost
- Enable Google Calendar API in APIs & Services > Library
- Add test users in OAuth consent screen > Audience (required while app is in Testing mode)

7. Key Technical Learnings

- Offline-first requires careful conflict resolution — UUID keys prevent integer collision between devices
- Real-time echo prevention — without markLocalWrite(), every write came back and toggled state twice
- Dexie schema migrations must define all previous versions or existing data is wiped
- Supabase free tier limits are real — magic link rate limit (3/hr) forced switch to password auth
- PWA installation requires locally-hosted PNG icons — external URL icons fail manifest validation

- Google OAuth popup flow and Supabase OAuth are completely separate — two different token systems
- Zustand `set()` vs `set(s => ...)` — must use functional form when reading previous state to avoid stale closures
- Vercel Edge Functions cannot access browser APIs but can access `process.env` — ideal for API proxying
- CSS custom properties (`var(--brand)`) allow theming without Tailwind config changes